Exam 3. Practice 2's answers keys.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

1. Solve each exponential equation.
   a. $5^x = 9$

      Answer: $x \approx 1.365$

   b. $4 \cdot 7^{-4x+1} - 1 = 2$

      Answer: $x \approx 0.287$

   c. $10^{4x+1} = 3^{-2x+3}$

      Answer: $x \approx 0.087$

2. A new artificial intelligence startup is valued at $2.5 million when it first launches in 2024. Market analysts predict the company's value will grow exponentially at a rate of 15% per year for the first 5 years.

   a. Write an equation to model the startup's valuation (V) in millions of dollars after t years.

      Answer: $V = 2.5(1.15)^t$

   b. Graph the equation
   c. According to the model, in what year will the company's valuation first exceed $10 million?

      ≈ 9.9 years.

Answer: Since $t \approx 9.9$, the valuation will exceed $10 million during the 10th year. If the start is 2024, this happens in **2034**.

   d. The founders decide that if the valuation reaches $15 million, they will sell the company. Use the model to estimate which year they will likely reach this milestone.

      Answer: They will likely reach this milestone in **2037** ($2024 + 13$ years).

3. A manufacturing plant purchases a precision robotic arm for $82,000. Because of wear and tear and the release of newer models, the plant estimates that the robot's value will decrease by 22% each year.

a. Write an equation to model the value of the robotic arm after t years.

Answer: $V = 82,000(0.78)^t$

b. Graph the equation
c. What is the estimated value of the robotic arm after 4 years?

Answer: The robotic arm will be worth approximately **$30,352.**

d. After how many years will the equipment be worth less than $15,000?

Answer: The value will drop below $15,000$ after approximately **6.84 years** (or during the 7th year).

4. A young couple decides to start a college savings fund for their child. They plan to deposit $4,000 at the end of each year for 18 years into an account that earns 5.2% interest compounded annually.

a. How much will be in the account after 18 years?

Answer: There will be approximately **$114,**652 in the account.

b. How much total interest was earned over the 18-year period?

Answer: The total interest earned is **$42,084.58.**

5. You have inherited a trust fund that will pay you $8,000 at the end of each year for the next 15 years. However, you have the option to take a single lump-sum payment today instead of waiting for the annual installments.

If the current market interest rate is 4.2% compounded annually, what is the present value of this trust fund today?

Final Answer

The present value of the trust fund today is approximately **$87,656.80**.

6. A person takes out a $30,000 loan to purchase a pre-owned vehicle. The credit union offers a fixed annual interest rate of 9% for a 60-month (5-year) term.

a. What can the buyer expect the monthly payment to be?

Answer: The monthly payment will be approximately **$622.75**.

b. How much will they pay in total for the car by the end of the 60 months?

Answer: The buyer will pay a total of **$37,365** for the car.

7. An entrepreneur takes out a $45,000 loan to launch a boutique coffee shop. The loan has an annual interest rate of 7.25%.

a. Find the monthly payment and calculate the total interest paid for a 3-year (36 months) repayment plan.

1. **Monthly Payment:**

$$M = 45000\frac{0.00604167(1+0.00604167)^{36}}{(1+0.00604167)^{36}-1} \approx \mathbf{\$1,394.51}$$

2. **Total Repayment:** $\$1,394.51 \times 36 = \$50,202.36$
3. **Total Interest Paid:**
   $\$50,202.36 - \$45,000 = \mathbf{\$5,202.36}$

b. Find the monthly payment and calculate the total interest paid for a 10-year (120 months) repayment plan.

1. **Monthly Payment:**

$$M = 45000 \frac{0.00604167(1 + 0.00604167)^{120}}{(1 + 0.00604167)^{120} - 1} \approx \mathbf{\$528.37}$$

2. **Total Repayment:** $\$528.37 \times 120 = \$63,404.40$

3. **Total Interest Paid:**
   $\$63,404.40 - \$45,000 = \mathbf{\$18,404.40}$